CORNELL TECH | APPLIED MACHINE LEARNING

Human or AI?

Style-aware text classification

Kaggle competition report | Group Number 11

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GitHub PR: https://github.com/sekhari/human-or-ai-submissions/pull/18

Working branch: `Group11retry`. This rebuild starts from the course repository main branch; it does not depend on the earlier team implementation.

Team information and model summary

Q0.1 *Team and contributions.*

Yilin Gao (yg695) led model development and comparison (Q1): data inspection, the shared split, representations, baseline comparisons and tuning. Ansh Raj Bhargava (ab3396) led the unlabeled-data and improvement experiments (Q3 and Q5). Yingzi Gu (yg667) led learning curves, subgroup evaluation and error analysis (Q2 and Q4). All three members also trained models and explored stronger configurations. These roles were supplied by the team; the implementation and experiment execution in this rebuilt branch were assisted by Codex.

Q0.2 *Final approach.*

The final detector combines 279 local writing-style features with 26 frozen GPT-2 language-predictability statistics. A 31-leaf LightGBM learns on supplied labels using original and whitespace-collapsed style views; the language features retain their original-text values in this feature-level augmentation. The final score is 75% of this new model plus 25% of the initial word/character/style ensemble. Selection uses the original fixed 2,000-row validation split, followed by refitting the tree models on all 10,000 labeled passages. Generic pretrained language weights run locally; no external authorship labels or hosted prediction APIs are used.

Q0.3 *Recorded scores.*

Metric	AUROC
Final fixed validation	0.999785
Initial public (earlier team model)	0.92433
First rebuild public	0.96962
Final optimized public (team-reported)	0.98665
Kaggle private	Not revealed

Interpretation. The public result comes from the team's Kaggle submission report. It is 0.013135 below validation. Repeated holdout selection and hidden-domain differences limit generalization claims; private performance remains unknown.

1. Model comparison and selection

Q1.1 *Preprocessing and representations.*

We decode literal backslash-n and backslash-t sequences into actual whitespace without discarding case, punctuation or paragraph boundaries. Word TF-IDF uses scikit-learn's default word token pattern, 1–2 grams, minimum document frequency 2, maximum document frequency 0.995, Unicode accent stripping, sublinear term frequency, L2 normalization and at most 180,000 features. Character TF-IDF uses 2–5 grams, minimum document frequency 3, sublinear term frequency, L2 normalization and the same feature cap. Our custom preprocessor preserves case, including for word features. Style features measure lexical diversity, hapax frequency, word/sentence lengths, punctuation, paragraph structure, function-word rates and several simple regex patterns. These represent complementary lexical and structural evidence. Each vocabulary and IDF transform is fitted on training texts only; row-local style extraction learns no corpus statistics.

Q1.2 *Model comparison on the same validation set.*

Model / features	Train AUC	Val. AUC	Seconds
Word LR, $C = 4$	1.000000	0.949536	2.51
Word linear SVM, $C = 1$	1.000000	0.952006	2.45
Character LR, $C = 16$	1.000000	0.977193	16.96
Word NB-weighted LR, $C = 4$	0.972718	0.938555	2.47
Style LightGBM, 31 leaves / 700 trees	1.000000	0.997234	5.75
Initial three-component blend	1.000000	0.997975	Shared fits

Times are local CPU feature-construction plus classifier-fit wall time, excluding scoring and checkpoint I/O; shared feature construction is attributed to each setting. They are approximate, not a hardware benchmark.

Q1.3 *Hyperparameters and selection protocol.*

Approach	Candidates	Selected
Word LR / SVM	$C \in \{0.1, 1, 4, 16\}$	SVM, $C = 1$
Character LR / SVM	$C \in \{0.1, 1, 4, 16\}$	LR, $C = 16$
NB-weighted LR (both representations)	$C \in \{0.1, 1, 4\}$	Not retained
Style LightGBM	Leaves $\{7, 15, 31\}$; trees $\{300, 700\}$	31 / 700

Seed 20261023 fixes a stratified 8,000/2,000 split. We compare 28 base settings and select the highest validation AUC within each representation. LightGBM uses learning rate 0.03, minimum child size 40, feature fraction 0.85 and L2 regularization 5. We evaluate 15 blend-grid entries: style weight $s \in \{0, .25, .5, .75, 1\}$, with remaining weight divided between word and character components using $c \in \{0, .5, 1\}$. Thus weights are $((1-s)(1-c), (1-s)c, s)$; pure-style endpoints repeat. No Kaggle feedback or test labels select settings. Reusing one validation set can still produce selection optimism.

Q1.4 *Final model choice and later changes.*

The initial three-component blend scored 0.997975 validation / 0.96962 public AUROC. A format-robust follow-up scored 0.995598 / 0.95899, so it was rejected despite strong synthetic stress results. Adding frozen language statistics then improved the selected blend to 0.999785 validation / 0.98665 public AUROC. The final setting uses original and collapsed style views, 31 leaves, 700 trees, learning rate 0.03 and L2 regularization 5. The unchanged baseline retains 25% weight as a conservative anchor. This was a later change motivated by measured public and validation evidence, not hidden individual labels. The appendix records the new search, pretrained source, grouped checks and compute cost; the initial comparison above is retained for context. Scores optimize ranking and are not calibrated authorship certainties.

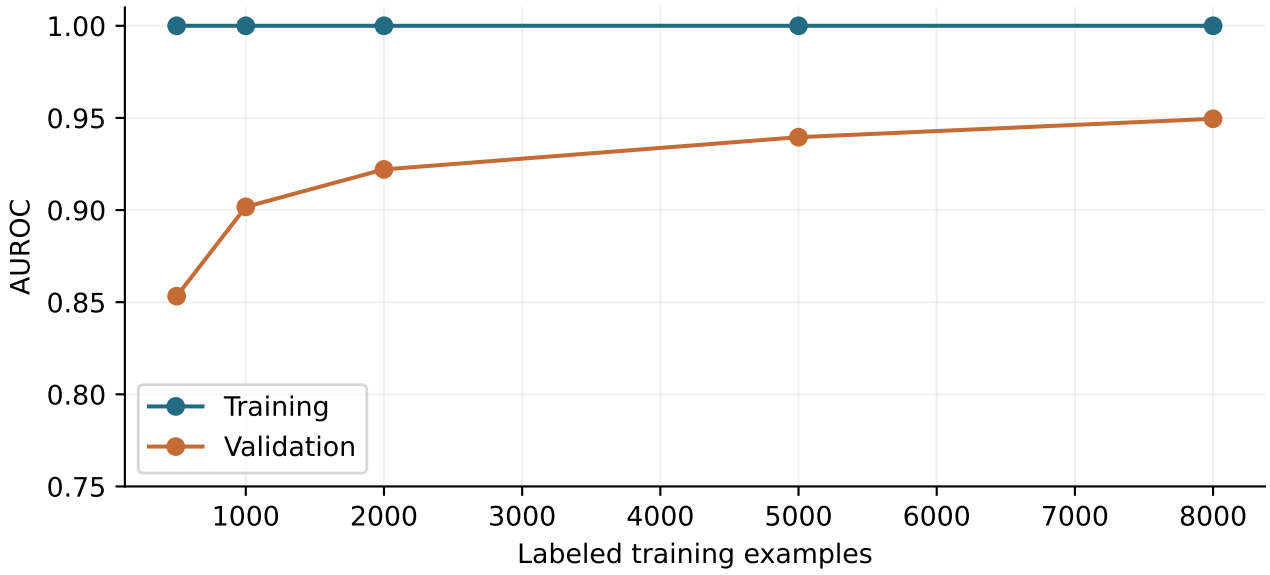
2. Effect of training data size

Q2.1 *Controlled learning curve.*

We fix word logistic regression at $C = 4$ and use stratified samples from the same 8,000-row training partition with seed 20261023. Each size gets a freshly fitted vocabulary and IDF, preventing smaller training sets from borrowing the full corpus representation. The 2,000 validation examples are unchanged. The smaller subsets are sampled separately; nesting is not required for this comparison.

Training rows	Training AUROC	Validation AUROC	Seconds
500	1.000000	0.853259	0.28
1000	1.000000	0.901718	0.41
2000	1.000000	0.922063	0.64
5000	1.000000	0.939543	1.40
8000	1.000000	0.949536	2.22

Q2.2 *Learning-curve figure.*



Q2.3 *Interpretation.*

Training AUC is 1.0 at every size, whereas validation AUC increases from 0.853259 at 500 examples to 0.949536 at 8,000. This is an overfitting pattern: a high-dimensional linear classifier separates its own examples even when the learned vocabulary and weights do not generalize as well. The increase from 5,000 to 8,000 examples is about 0.0100, so additional labeled data remains useful for this configuration. The curve does not show substantial training underfitting and does not establish that this model has saturated. Only one sample per size is measured; the curve has sampling variability, and no asymptotic rate is inferred. It is a learning curve for word LR, not evidence that the final style-heavy blend would improve by the same amount.

3. Learning from unlabeled data

Q3.1 *Method and leakage prevention.*

We test unsupervised representation learning using the additional 3,000 unlabeled texts. The supervised baseline fits word vocabulary and IDF only on the 8,000 training texts. The alternate representation fits vocabulary and IDF on these same texts plus all 3,000 unlabeled texts. Both classifiers use logistic regression with $C = 4$ and train on exactly the same 8,000 true labels. Neither validation texts nor validation labels enter either representation fit, and test texts are excluded. This experiment does not create pseudo-labels or assume unlabeled authorship. The fixed holdout isolates the effect of changing corpus statistics, although it cannot establish whether the same change would help hidden test domains.

Q3.2 *Measured comparison.*

Method	Validation AUROC
Supervised word LR only	0.949536
Training + unlabeled vocabulary/IDF	0.949414

Q3.3 *Did unlabeled data help?*

No improvement was observed: validation AUC changes by -0.000122 . This difference is tiny and should be interpreted as essentially neutral rather than strong evidence of harm. Additional documents change vocabulary coverage and IDF weights but supply no new class information. They may add domain vocabulary that is not useful for authorship discrimination; this is a hypothesis, not a measured causal explanation. Pseudo-label noise cannot explain this result because no pseudo-labels were used. The baseline component therefore retains its supervised-only word representation fit. A future self-training experiment would need a fixed confidence rule, comparison on the same holdout, and explicit checks for confirmation bias rather than an assumption that more unlabeled data always helps. **Data accounting.** The starter file is a subset of the supplied labeled set and is never appended as

additional training data. The 3,000 test texts are used only for inference. No hidden labels are recovered from the dataset's external source or from external model services.

4. Generalization and error analysis

Q4.1 *Final training–validation gap and uncertainty.*

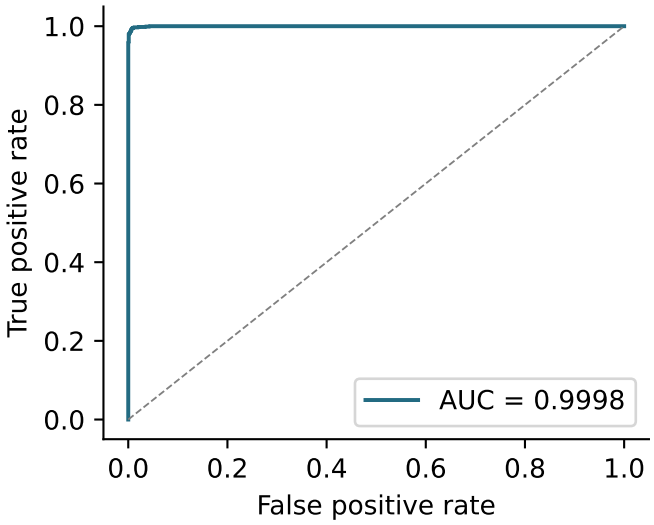
The final selected model has training AUC 1.000000 and validation AUC 0.999785. These scores use the controlled 8,000/2,000 fit before full-data refitting. A 1,000-replicate stratified bootstrap gives [0.999595, 0.999924], conditional on the selected fitted model; it excludes selection and distribution-shift uncertainty. No exact texts cross the split. Two validation passages have nearest-training word-TF-IDF cosine similarity at least 0.90; removing them yields AUC 0.999785. This is not a complete shared-source audit. The public AUC is lower than random-split AUC. That observed gap, and the failed robust candidate, caution against treating a near-perfect holdout as a universal authorship detector.

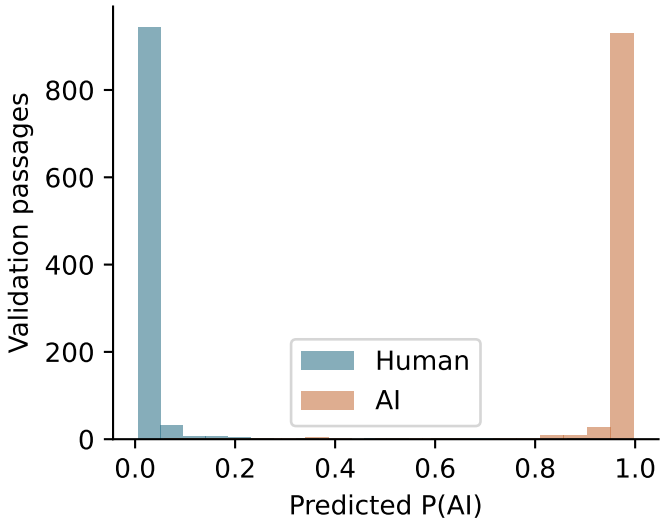
Q4.2 *Final observable text-length subgroups.*

We count whitespace-delimited words after decoding escaped newlines and keep the original bins, independent of labels and model predictions.

Words	Count	Human / AI	AUROC
0 <= words < 150	380	146 / 234	0.999766
150 <= words < 250	1049	605 / 444	0.999773
words >= 250	571	249 / 322	0.999888

All three bins have AUC near 1.0. Unequal group sizes and tiny differences do not establish a reliable subgroup advantage. These length groups do not test new generators or domains; the appendix adds unsupervised topic/style-cluster checks, which are also only proxies.





Q4.3 *Final errors at threshold 0.5.*

The confusion matrix is TN=995, FP=5, FN=13, TP=987, with accuracy 0.9910. A false positive is human text scored at least 0.5; a false negative is AI text scored below 0.5. The next page shows the five highest-scoring false positives and five lowest-scoring false negatives. These deliberately extreme cases are not a representative error sample.

Generalization: concrete errors

The observed content descriptions below come from reviewing the supplied validation passages. Proposed explanations are hypotheses; we did not intervene on each passage to establish causation.

ID	Type	P(AI)	Observed content and possible failure mechanism
2696	FP	0.9095	Human-labeled forum text repeats long phrases extensively; repetition is a plausible misleading generation cue.
9585	FP	0.6964	Fluent encyclopedia biography with repeated names, dates and media titles resembles generated reference prose.
6138	FP	0.6805	Formal historical description with many named entities overlaps the style of generated summaries.
7384	FP	0.6012	Very short film review with broad judgments; limited text makes style estimates uncertain.
6404	FP	0.5049	Informal first-person self-report with unusual spelling and contractions; its score is close to the threshold.
7049	FN	0.0578	Recipe quantities, bullets and personal restaurant commentary resemble human recipe/blog material.
3457	FN	0.0689	Quoted, line-wrapped academic abstract looks like source material; formatting and vocabulary may mask authorship cues.
9298	FN	0.1134	Biography-like prose with dates, names and irregular spacing resembles human reference writing.
3061	FN	0.1965	Quoted parenting discussion uses informal, personal language that resembles human forum text.
1689	FN	0.2014	Dense mathematical notation and code fences are poorly represented by English predictability statistics.

Q4.4 *Recurring final-model failures.*

The five human errors include repetitive forum language, two reference-style descriptions, a short review and an informal self-report. AI misses include a recipe, an academic abstract, a biography, parenting discussion and symbolic mathematics. Thus neither fluent reference prose nor conversational or technical irregularity uniquely identifies authorship. The highly repetitive human example also shows why a predictability signal cannot establish generation on its own. These are observations about the selected examples; the proposed mechanisms are hypotheses, not causal tests or recovered source/generator identities. Additional training-only symbolic-text augmentation or a representation better suited to mathematics could be tested next. Language features reduced total threshold errors from 31 to 18, but some initial errors remain, and the public subset still performs below the matched validation pool.

5. Open-ended improvements and insights

Q5.1 *Hypothesis.*

Lexical models and style features should make different errors, so their blend may improve ranking over either representation alone. The fixed-holdout results support this as a useful experiment: the character model is substantially weaker than style boosting but can still contribute complementary information. After the blend experiment, feature-gain inspection showed paragraph-format dependence. We therefore also tested whether whitespace augmentation could preserve native-text AUC while improving robustness to a formatting shift. This second question matters because random splits retain acquisition and formatting regularities that may not survive different preprocessing or adversarial edits.

Q5.2 *Exact changes and controlled comparison.*

For the blend improvement, we keep the labeled split and trained components fixed and change only the mixture weights, using the finite grid described in Q1.3. For the augmentation follow-up, each training passage contributes its original and whitespace-collapsed version with the same supplied label. The 8,000 source training passages become 16,000 transformed training rows, not additional independent labeled examples. We compare 15- and 31-leaf style models with 700 trees and style weights 0.75 or 1.0; lexical components stay fixed. Before executing this follow-up, we specified an acceptance rule: native AUC must be within 0.0005 of the original blend and collapsed-text AUC must be at least 0.98. Eligible candidates would be ranked by native AUC.

Q5.3 *Measured results.*

Configuration	Native AUC	Collapsed AUC
Character LR alone	0.977193	Not isolated
Style boosting alone	0.997234	Not isolated
Initial three-component blend	0.997975	0.877512
Augmented 31-leaf style + lexical blend	0.997503	0.978602

The selected blend reuses the component fits from Q1.2; prediction adds three component passes. The 31-leaf augmented candidate takes 6.57 seconds for feature construction and fitting with cached original features; this excludes the already fitted lexical components and is not a full-pipeline speed comparison. Removing size and sentence/paragraph-count features from the original style model yields AUC 0.996225 (baseline 0.997234). This partial ablation still retains newline-character frequency, so it does not establish formatting independence.

Q5.4 *What we learned and limitations.*

Blending gives a small native-text ranking improvement, but it does not solve formatting dependence. Whitespace augmentation raises collapsed-text AUC from 0.877512 to 0.978602 for the strongest native-text augmented candidate, while reducing native AUC from 0.997975 to 0.997503. No candidate passes the prespecified 0.98 collapsed-text requirement, so augmentation alone was rejected; the later language-feature model is evaluated in the appendix. ASCII punctuation normalization retains AUC 0.997797, and bracket-reference removal retains 0.997425; whitespace is the clearly larger measured vulnerability. The three frozen-setting secondary random splits score 0.998941, 0.997419, 0.998878 . They support random-split stability but overlap the selection pool and do not constitute external validation. These results support the narrow claim that the model ranks supplied unmodified validation passages well. They do not establish robustness to hidden adversarial transformations or a specific Kaggle score. A next experiment would canonicalize formatting throughout training and inference and compare its worst-transformation AUC under a separately reserved evaluation protocol.

6. Code, reproducibility and AI assistance

Q6.1 *PR, environment and entrypoints.*

PR URL: https://github.com/sekhari/human-or-ai-submissions/pull/18

Location: `submissions/group-number-11/`.

Initial CPU experiments used macOS arm64 and Python 3.14.2. Optimization added local Apple MPS feature extraction; exact added packages are in requirements-optimization.txt. Exact package versions are recorded in `environment.txt`; install dependencies from `requirements.txt`. Main split and model seed: 20261023. Secondary split seeds: 20261024, 20261025, 20261026. LightGBM has deterministic mode enabled; small numeric differences across platforms remain possible.

```
DATA_DIR=cornell-tech-cs-5785-combined-2026-ai-vs-human-text
bash submissions/group-number-11/reproduce_optimized.sh "$DATA_DIR"
python submissions/group-number-11/predict_optimized.py \
  --test "$DATA_DIR/test.csv" \
  --sample "$DATA_DIR/sample_submission.csv" \
  --model submissions/group-number-11/artifacts/optimization/language_v2.joblib \
  --out submissions/group-number-11/output/submission_language_v2.csv
```

The `reproduce_optimized.sh` entrypoint runs baseline fitting, frozen language-feature extraction, classifier search, final refitting and prediction. The README documents supplementary diagnostics and report generation. Data, model checkpoints and intermediate feature caches remain local and are excluded from Git.

Q6.2 *Development history and lessons.*

- e31a805: Build fixed-holdout text and stylometric training pipeline
- cd956bc: Audit generalization and test whitespace augmentation for Group Number 11
- 675a469: Record controlled experiments and deliver reproducible report and submission workflow

The initial pipeline established one reusable holdout and showed that style features outperform sparse lexical baselines on this data. The generalization audit showed that the score survives removing two high-similarity validation examples but fails a whitespace-collapse stress test. The augmentation follow-up measured a substantial robustness gain that still missed its predefined acceptance criterion. These lessons are recorded as actual experiments rather than retrospective claims that every change helped. Commit hashes identify development history; inspect the hosted PR or use `git show HASH` locally.

Q6.3 *AI assistance and verification.*

OpenAI Codex assisted with pipeline design, code, debugging, experiment execution, analysis and report drafting. Experiments were run locally on the supplied data; offline scores come from their logs, while the public score was supplied by the team. Automated checks verify the fixed split, validation labels/AUC, data hashes, output IDs/order and finite probability bounds. No hosted LLM API labeled or predicted passages. Optimization used generic GPT-2 pretrained weights locally; additional neural experiments are recorded in the optimization logs. No external authorship labels were used. The team supplied the individual roles recorded in Q0.1, and reports that all three members also trained models and explored stronger configurations. This describes the team's reported division of work, while the implementation and experiment execution in this rebuilt branch were assisted by Codex. Before course submission, the team needs to review the code, interpretations and latest leaderboard result. **Sources.** The supplied *Kaggle Competition Template Report* PDF/LaTeX provides the protocol and Q0–Q6 requirements; the course README provides the repository layout. Offline results come from saved experiment logs; public results were supplied by the team. No external label source was consulted. **Submission status.** The team reported public AUROC 0.98665 for the optimized submission, following 0.96962 for the first rebuild and 0.92433 for the earlier model. Private results remain pending. See the title page for the draft PR.

Appendix: post-submission optimization (Q1/Q5/Q6 continuation)

Measured sequence and selection. The first format-robust candidate improved synthetic stress AUC but reduced public AUC, an informative negative result. The language-feature candidate was then selected using the original fixed holdout: retain at least 25% baseline weight, require native AUC at least 0.9975, and maximize 0.75 times the no-blank-line subgroup AUC plus 0.25 times overall AUC. This focuses on passages without an easy paragraph cue. A finite grid compared baseline weights 0.25, 0.5 and 0.75, with plain or conditional-rank blending. Plain blending won; inference does not use ranks or true labels.

Configuration	Validation AUC	Public AUC
First rebuild ensemble	0.997975	0.96962
Robust format candidate	0.995598	0.95899
GPT-2 statistics + baseline	0.999785	0.98665

Representation and search. The additional 279-dimensional representation includes the original 160 style features plus lexical-diversity, n-gram repetition, word/sentence distributions and character/paragraph statistics. We concatenate 26 frozen GPT-2 features: token negative-log-likelihood distribution statistics, head/tail-window statistics, rank-1/10/100 fractions and logit-margin statistics. Only the first 256 model tokens are used. The generic model is openai-community/gpt2, revision `607a30d783dfa663caf39e06633721c8d4cfcd7e`; its pretrained weights are unchanged. No competition passage is sent to a hosted endpoint. The new grid compares LightGBM leaves 7/15/31 and CatBoost depth 5 in native, collapsed and paired native/collapsed style views. Language features retain original text in those paired style views, so style-only stress scores are not reported as full-pipeline format robustness.

Additional grouped evidence. Twelve unsupervised lexical topic/style clusters define three group-held-out folds, without hidden source labels. Frozen style-only and style-plus-GPT-2 configurations use their separately selected 31- and 15-leaf settings. This is a pipeline comparison, not an isolated feature ablation:

Fold	Style only	Style + GPT-2
1	0.994101	0.999002
2	0.998777	0.998415
3	0.990991	0.997209

The small regression on fold 2 cautions against universal improvement claims. These clusters are proxies, and their folds overlap the original selection pool; they are not independent external confirmation.

Compute and limitations. Language-feature extraction took minutes on the local Apple GPU and is cached for reuse; the selected classifier stage took about 23 seconds with cached features. This is not directly comparable to the initial CPU feature-plus-fit timings. The final tree fits use all 10,000 true labeled rows after selection, with paired style views counted as augmentation rather than new independent data. Generic pretrained language knowledge is the only added external model resource. Public feedback was used to reject the robust candidate and confirm the language candidate; no hidden individual labels were accessed. The repeated holdout search and small public subset remain sources of selection uncertainty.